PROBLEM SET 7, PART 2: DE RHAM COHOMOLOGY GROUPS
DUE: DEC. 28

(1) [**The de Rham cohomology groups of punctured manifolds**]
   (a) Compute $H^1_{dR}(\mathbb{T}^2 - \{p\})$.
   (b) Suppose $n > 1$. Compute $H^k_{dR}(\mathbb{R}^n - \{p, q\})$ and $H^k_c(\mathbb{R}^n - \{p, q\})$.
   (c) Let M be a compact oriented connected smooth manifold of dimension $m > 1$, $p \in M$.
      (i) Prove: $H^m_{dR}(M \setminus \{p\}) = 0$.
      (ii) For $k < m$, prove: $H^k_{dR}(M \setminus \{p\}) = H^k_{dR}(M)$.

(2) [**Invariance of compactly supported de Rham cohomology groups**]
Prove Theorem 1.3.3 and corollary 1.3.5.

(3) [**The mapping degree**]
   (a) Let $f : S^1 \to S^1$ be the map $f(\theta) = e^{ik\theta}$. Find $\deg(f)$.
   (b) Let $f : \mathbb{T}^n \to \mathbb{T}^n$ be given by $f(e^{i\theta_1}, \cdots, e^{i\theta_n}) = (e^{ik_1\theta_1}, \cdots, e^{ik_n\theta_n})$. Find $\deg(f)$.
   (c) Prove: If $f : S^m \to S^m$ has degree $\deg(f) \neq (-1)^m$, then it admits at least one fixed point.
   (d) Prove: Any smooth map $f : S^2 \to \mathbb{T}^2$ has degree 0.

(4) [**The fundamental theorem of algebra**]
   (a) Let $f : \mathbb{C} \to \mathbb{C}$ be a polynomial of degree $k > 0$. Prove: f is proper, and is properly homotopic to $f_1(z) = z^k$.
   (b) Use mapping degree to prove the fundamental theorem of algebra.

(5) [**Orientation covering**] (NOT required)
Let M be a connected non-orientable manifold of dimension m. Let

$$\widetilde{M} = \{(p, O_p) \mid p \in M, O_p \text{ is an orientation on } T_pM\}.$$

Let $\pi : \widetilde{M} \to M$ be the obvious projection map.
   (a) Show that $\widetilde{M}$ admits a natural topological and smooth structure so that it is an m-dimensional connected orientable manifold, and π is a smooth double covering map.
   (b) Show that the non-trivial Deck transformation $\sigma : \widetilde{M} \to \widetilde{M}$ is an orientation-reverting diffeomorphism.
   (c) Find the relation between $H^k_{dR}(M)$ and $H^k_{dR}(\widetilde{M})$.

(6) [**The Euler characteristics**]

(a) Let B be a compact smooth manifold with boundary ∂B. Let $2B$ be the manifold obtained by "gluing" two copies of B along the boundary. Prove:

$$\chi(2B) + \chi(\partial B) = 2\chi(B).$$

(b) (NOT required) Given the following formula (the answer to a problem in previous PSet)

$$H^k_{dR}(\mathbb{RP}^{2n}) = \begin{cases} \mathbb{R}, & k = 0, \\ 0, & k \neq 0. \end{cases}$$

Prove: $\mathbb{RP}^{2n}$ can not be the boundary of any $2n+1$ dimensional compact manifold.

(7) [H^1_{dR} **v.s.** π_1] (NOT required)

Let M be a compact connected smooth manifold. Consider the map

$$\Psi : H^1_{dR}(M) \to \mathrm{Hom}(\pi_1(M), \mathbb{R}), \qquad [\omega] \mapsto \left([\gamma] \mapsto \int_\gamma \iota_\gamma^* \alpha \right)$$

where we only take smooth representatives $\gamma \in [\gamma]$, and the map $\iota_\gamma : \gamma \hookrightarrow M$ is the inclusion map.

(a) Prove: The map Ψ is well-defined.
[Hint: use the cochain homotopy property.]

(b) Prove: The map Ψ is injective.
[In fact the map Ψ is also surjective, but the proof is hard.]
In particular, if M is simply connected, then $H^1_{dR}(M) = 0$.

(8) [**Moser's trick**] (NOT required)

Let M be a m-dimensional compact orientable manifold, and $\omega_0, \omega_1 \in \Omega^m_{dR}(M)$ are two volume forms on M such that $\int_M \omega_0 = \int_M \omega_1$. Show that there exists a diffeomorphism $\varphi : M \to M$ such that $\varphi^* \omega_2 = \omega_1$ as follows:

(a) For $t \in [0,1]$, let $\omega_t = (1-t)\omega_1 + t\omega_2$. Prove: ω_t is a family of volume forms on M, and there exists $\alpha \in \Omega^{m-1}(M)$ so that $\dot{\omega}_t = d\eta$ holds for all $t \in [0,1]$.

(b) Let X_t be the time-depending vector field so that $\iota_{X_t}\omega_t = -\alpha$. Let φ_t be the flow of X_t. Prove: $\varphi_t^* \omega_t = \omega_0$ holds for all $t \in [0,1]$.